Time-Consistent Policy with Pre-Tax or Post-Tax Retirement Accounts

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Defined Contribution Plans in the United States

- A very large amount of wealth:
  - Total U.S. retirement assets $\approx$ \$45.8 trillion (mid-2025)
  - IRAs + employer DC plans $\approx$ 67% of all retirement assets
  - DC/IRA wealth $\approx$ 105% of U.S. GDP
- Also a large fraction of household wealth:
  - Retirement accounts are among the largest asset classes for middle-income households
  - 44% of U.S. households own an IRA; many also hold 401(k)s
- Retirement wealth is highly concentrated:
  - Median balance (among account holders): \$87,000
  - Mean balance: \$334,097 — nearly 4$\times$ the median
  - In 2021, about 500 accounts worth \$25 million or more (highest around \$5 billion)
  - Bottom 46% of households: *no* DC or IRA savings at all

Roth vs. Traditional Plans

- Plans come in two flavors:
  - Traditional (pre-tax): Contributions are tax-deductible; withdrawals taxed as ordinary income
  - Roth (after-tax): Contributions made from after-tax dollars; withdrawals entirely tax-free
  - Both avoid double taxation of returns (gains accrue tax-deferred or tax-free)
- Brief history:
  - Traditional IRAs: introduced 1974 (ERISA); 401(k)s: 1978
  - Roth IRAs: introduced 1998; Roth 401(k)s: available 2006
  - Gradual expansion: Roth first in IRAs, then 401(k)s, now mandatory for catch-up contributions (high earners, SECURE 2.0, 2022)

The Current Landscape: Stock and Flows

- **Stock**: Traditional accounts overwhelmingly dominate
  - Traditional IRAs $\approx$ 84% of IRA assets; Roth IRAs $\approx$ 10%
  - (Remainder in employer-sponsored IRAs, no breakdown for 401ks)
- **New saving**: Roth is competitive
  - Direct IRA contributions (2022): $\sim$10 million Roth contributors vs. $\sim$5 million traditional
  - Aggregate: \$35 bn to Roth vs. \$22 bn to traditional
  - Roth 401(k) participation: 18% of eligible workers (2024 all-time high, up from 12% in 2019)
- **Conversions**: $\sim$\$35 bn/year
- $\Rightarrow$ Roth is a growing share of retirement saving; policy question is increasingly important

Just an Accounting Gimmick?

- Suppose constant income tax τ, constant real rate r, horizon T
- Households:
  - Save \$1 in traditional $\Rightarrow$ receive \$ $(1+r)^T(1-\tau)$ upon retirement
  - Save \$ $(1-\tau)$ in Roth + pay \$ τ in taxes $\Rightarrow$ receive \$ $(1+r)^T(1-\tau)$
  - *Same after-tax return*
- Government:
  - Traditional: collects $\tau(1+r)^T$ taxes at retirement
  - Roth: collects τ taxes today
  - Present value is the same
  - The only difference: Roth books the revenue early $\Rightarrow$ lower reported deficit today
- $\Rightarrow$ Under these assumptions, no real difference

When Roth Looks Better: Landoni and Zeldes (2025 RFS)

- Suppose households pay an intermediation cost δ on their savings
- Households:
  - Save \$1 in traditional $\Rightarrow$ \$ $[(1+r)(1-\delta)]^T(1-\tau)$
  - Save \$ $(1-\tau)$ in Roth $\Rightarrow$ \$ $[(1+r)(1-\delta)]^T(1-\tau)$
  - Still the same!
- But the government now collects:
  - Traditional: taxes $\tau[(1+r)(1-\delta)]^T$ at retirement; PV $= \tau(1-\delta)^T$
  - Roth: taxes τ today;
- $\Rightarrow$ If savings face a cost, traditional *shrinks* the tax base.

Our Take: Two Major Problems with Roth

1. Lack of Intergenerational Risk Sharing
   - Roth pre-commits that the old will not contribute to financing future government spending
   - This is a commitment to make insurance markets incomplete

2. Time Inconsistency
   - "Trust me, we will never tax this money again"
   - But in period 2, taxing the old's retirement accounts is *lump sum* ex post
   - Pool of untaxed money in the hand of the rich
   - The government has a strong incentive to renege
   - *Roth strengthens the temptation to deviate from past promises*

Plan of the Talk

- Simple two-period, two-generation analytical model
  - Characterize competitive equilibria under Roth and traditional 401k
  - Compare Ramsey allocations
  - Analyze time inconsistency: which regime requires stronger commitment?
- Quantitative OLG model (setup only – results forthcoming)
  - N-period-lived households with realistic lifecycle earnings
  - Contribution limits, progressive taxation, Social Security
  - Political economy: voting on tax rates after an unexpected spending shock

Environment

- Two periods, two generations of size 1
  - Generation 1: lives in periods 1 and 2
  - Generation 2: lives in period 2 only
- Each agent endowed with 1 unit of time (labor = output)
- Generation 1 can save via storage at gross return $(1 + r)$
- Government must finance a stochastic expenditure g_2 in period 2, drawn from distribution G

Preferences and Market Structure

- Generation 1:

$$u(c_{1y}) - v(\ell_1) + \mathbb{E}[u(c_{2o})]$$

- Generation 2:

$$u(c_{2y}) - v(\ell_2)$$

- In period 1, generation 1 can also trade *state-contingent* bonds at price q_2 (in zero net supply unless government participates)
- Two tax regimes:
  - **Roth**: tax labor in both periods; promise no tax on savings in period 2
  - **Traditional 401k**: exempt savings from period-1 tax; tax withdrawals at rate τ_2^s in period 2

Competitive Equilibrium: Roth

- Budget constraints:

$$c_{1y} \leq \ell_1(1-\tau_1^\ell) - s_1 - \mathbb{E}[q_2 b_2]$$

$$c_{2o} \leq s_1(1+r) + b_2, \qquad c_{2y} \leq \ell_2(1-\tau_2^\ell)$$

- Key optimality conditions:

$$u'(c_{1y}) = (1+r)\,\mathbb{E}[u'(c_{2o})], \qquad q_2\, u'(c_{1y}) = u'(c_{2o})$$

Primal Characterization: Roth

$(c_{1y}, c_{2o}, c_{2y}, \ell_1, \ell_2)$ is a competitive equilibrium $\iff$

1. Market clearing for goods (periods 1 and 2)
2. Euler equation: $u'(c_{1y}) = (1+r)\mathbb{E}[u'(c_{2o})]$ [*extra constraint vs. traditional*]
3. Cohort 1 implementability: $u'(c_{1y})c_{1y} + \mathbb{E}[u'(c_{2o})c_{2o}] = v'(\ell_1)\ell_1$
4. Cohort 2 implementability: $u'(c_{2y})c_{2y} = v'(\ell_2)\ell_2$

Extra restriction (no state-contingent government bonds):

- c_{2o} must be *independent* of g_2
- The Roth promise leaves the first generation fully insulated from fiscal shocks

Competitive Equilibrium: Traditional 401k

- Budget constraints:

$$c_{1y} \leq (\ell_1 - s_1 - \mathbb{E}[q_2 b_2])(1 - \tau_1^\ell)$$

$$c_{2o} \leq (s_1(1+r) + b_2)(1 - \tau_2^s), \qquad c_{2y} \leq \ell_2(1 - \tau_2^\ell)$$

- Key optimality condition:

$$(1 - \tau_1^\ell) u'(c_{1y}) = (1+r)\,\mathbb{E}[(1 - \tau_2^s) u'(c_{2o})]$$

- Could add restriction $\tau_1^\ell = \tau_2^s$...
- ...then traditional 401k may be restrictive, but:
  - Simple model does not have heterogeneity within cohort
  - In practice there are other age-based systems for redistribution across cohorts

Primal Characterization:Traditional

- same implementability constraints as Roth, but no extra Euler equation constraint
- τ_2^s can be state contingent, no need for state-contingent debt
  - Government can raise τ_2^s in bad states $\Rightarrow$ shares risk with old generation
  - Even without state-contingent bonds, full risk sharing is achievable

Roth as a Commitment to Market Incompleteness

- Under Roth (without government state-contingent bonds):
  - c_{2o} must be *non-contingent* on the realization of g_2
  - The first generation is fully insulated from fiscal shocks
  - The entire adjustment falls on the second generation
- Under regular 401k:
  - The tax τ_2^s on withdrawals can be state contingent
  - The old can be asked to contribute to financing g_2
  - Even without state-contingent bonds, full risk sharing is achievable in this model
- Interpretation: Roth is not just a tax timing change. It is a commitment to a particular *distribution of risk* between generations.

Ramsey Allocations

- Government maximizes weighted social welfare:

$$\omega \left\{ u(c_{1y}) - v(\ell_1) + \mathbb{E}[u(c_{2o})] \right\} + (1-\omega)\, \mathbb{E}\left[u(c_{2y}) - v(\ell_2) \right]$$

  subject to constraints from the primal characterization
- Under traditional 401k, FOCs for c_{2o} and c_{2y}:

$$\left[\omega + \lambda_1(1-\sigma)\right] c_{2o}^{-\sigma} = \mu_2$$

$$\left[1 - \omega + \lambda_2(1-\sigma)\right] c_{2y}^{-\sigma} = \mu_2$$

  (λ_1, λ_2: Lagrange multipliers on implementability constraints)
- Under Roth: additional constraint $u'(c_{1y}) = (1+r)\mathbb{E}[u'(c_{2o})] \Rightarrow$ constrained Ramsey problem with a smaller feasible set
- $\Rightarrow$ Ramsey welfare under Roth $\leq$ Ramsey welfare under regular 401k

Analytical Example: Setup

- Specialize to $u(c) = c^{1-\sigma}/(1-\sigma)$, general $v(\ell)$
- Benchmark: $g_2 = 0$ and laissez-faire is optimal
  - Pareto weights satisfy $\omega/(1-\omega) = u'(c^*_{2o})/u'(c^*_{2y})$
  - Denote laissez-faire allocation with *; at this point Lagrange multipliers on implementability are zero
  - For these preferences, Euler constraint does not bind, restriction on Roth comes from lack of state-contingent bonds
- Perturbation: $g_2 = \alpha G_2$ with α small; study the Ramsey allocation to first order in α
- Notation:
  - $\hat{x}$ = log-deviation of x from its laissez-faire value
  - $\tilde{x}$ = level-deviation of x from its laissez-faire value

Linearized Ramsey Conditions

- Market clearing (linearized):

$$c_{1y}^* \hat{c}_{1y} + s_1^* \hat{s}_1 = \ell_1^* \hat{\ell}_1$$

$$c_{2o}^* \hat{c}_{2o} + c_{2y}^* \hat{c}_{2y} + \tilde{g}_2 = s_1^*(1+r)\hat{s}_1 + \ell_2^* \hat{\ell}_2$$

- Implementability constraints (linearized):

$$(1-\sigma)\left[c_{1y}^* \hat{c}_{1y} + (1+r)c_{2o}^* \mathbb{E}\hat{c}_{2o}\right] = (1+\epsilon_v(\ell_1^*))\ell_1^* \hat{\ell}_1$$

$$(1-\sigma)\hat{c}_{2y} = (1+\epsilon_v(\ell_2^*))\hat{\ell}_2$$

- First-order conditions for c_{2o} and c_{2y}:

$$-\omega\sigma\hat{c}_{2o} + (1-\sigma)\tilde{\lambda}_1 = \omega\hat{\mu}_2$$

$$-(1-\omega)\sigma\hat{c}_{2y} + (1-\sigma)\tilde{\lambda}_2 = (1-\omega)\hat{\mu}_2$$

where μ_2 is the shadow cost of period-2 resource feasibility and λ_1, λ_2 are the multipliers on the implementability constraints

Differences between traditional and Roth

- To first order, period-1 decisions are identical across Roth and traditional 401k
  - Both depend only on $\mathbb{E}\tilde{g}_2$; differences arise only through $\tilde{g}_2 - \mathbb{E}\tilde{g}_2$

Solving the System: traditional 401k

- The system is linear $\Rightarrow$ decompose into:
  - *Mean* component: depends on $\mathbb{E}\tilde{g}_2$ (same across regimes)
  - *Deviation* component: depends on $\tilde{g}_2 - \mathbb{E}\tilde{g}_2$ (differs across regimes)
- Focus on deviations. For traditional 401k, solving gives:

$$\hat{\mu}_2 - \mathbb{E}\hat{\mu}_2 = \frac{(\tilde{g}_2 - \mathbb{E}\tilde{g}_2)(1 + \sigma\epsilon_v(\ell_2^*))}{c_{2o}^*/\sigma + c_{2y}^*(\epsilon_v(\ell_2^*) + \sigma)}$$

- Welfare loss from spending volatility (second order):

$$\mathcal{L}^{\text{reg}} = \text{Var}(g_2) \cdot \frac{1 + \sigma\epsilon_v(\ell_2^*)}{c_{2o}^*/\sigma + c_{2y}^*(\epsilon_v(\ell_2^*) + \sigma)}$$

- Intuition: higher σ or higher c_{2o}^* (more savings to tax) reduces the marginal cost of responding to g_2, since the government can spread the burden across both generations

Solving the System: Roth

- Under Roth, the constraint $\hat{c}_{2o} = \mathbb{E}\hat{c}_{2o}$ is imposed
  - The first generation's old-age consumption cannot respond to g_2
  - The entire period-2 adjustment must fall on c_{2y} and ℓ_2
- The shadow cost of resources now satisfies:

$$\hat{\mu}_2 - \mathbb{E}\hat{\mu}_2 = \frac{(\tilde{g}_2 - \mathbb{E}\tilde{g}_2)(1 + \sigma\epsilon_v(\ell_2^*))}{c_{2y}^*(\epsilon_v(\ell_2^*) + \sigma)}$$

- Welfare loss from spending volatility (second order):

$$\mathcal{L}^{\text{Roth}} = \text{Var}(g_2) \cdot \frac{1 + \sigma\epsilon_v(\ell_2^*)}{c_{2y}^*(\epsilon_v(\ell_2^*) + \sigma)}$$

Comparing the Two Welfare Losses

- The denominators tell the story directly:

$$\underbrace{c_{2o}^*/\sigma + c_{2y}^*(\epsilon_v(\ell_2^*) + \sigma)}_{\text{traditional 401k denominator}} \quad > \quad \underbrace{c_{2y}^*(\epsilon_v(\ell_2^*) + \sigma)}_{\text{Roth denominator}}$$

- Therefore $\mathcal{L}^{\text{Roth}} > \mathcal{L}^{\text{reg}}$ always
- The missing term c_{2o}^*/σ in the Roth denominator is precisely the cost of the market incompleteness:
  - Under traditional 401k, a fraction of the burden of $\tilde{g}_2$ is borne by the first generation through c_{2o}
  - The larger c_{2o}^* (more retirement savings), the greater the welfare cost of the Roth commitment
  - The larger σ (more curvature, more insurance value), the more costly the restriction
- The welfare gap grows with the stock of retirement savings and with risk aversion

Time Inconsistency: Setup

- In period 2, the government can reoptimize, taking s_1 and B_2 as given
- Under reoptimization:
  - The old's budget constraint is $c_{2o} = (s_1(1+r) + B_2)(1 - \hat{\tau}_2^s)$
  - Choosing $\hat{\tau}_2^s$ is equivalent to choosing c_{2o} directly
  - The tax on the old is lump sum ex post — no distortion at the margin
  - No implementability constraint for generation 1 ($\lambda_1 = 0$)
- The reoptimized FOC for c_{2o} is:

$$\omega\, c_{2o}^{-\sigma} = \mu_2^{\text{Reopt}}$$

  identical to the Ramsey FOC but with $\lambda_1 = 0$
- Whenever $\lambda_1 > 0$ in the original Ramsey plan, reoptimization is tempting

Time Inconsistency: The Minimum Commitment Cost

- Define M^* as the minimum utility cost of reneging that deters deviation
- Proposition: to second order,

$$M^* = \max_{\tilde{g}_2} \Big[\underbrace{(\tilde{\mu}_2^{\text{Ramsey}} - \tilde{\mu}_2^{\text{Reopt}})\tilde{g}_2}_{\text{gain from better resource sharing}} + \underbrace{\tilde{\lambda}_1 \tilde{c}_{2o}^{\text{Ramsey}}}_{\text{gain from lump sum tax}} \Big]$$

- The first term: the reoptimized plan gives the second generation more relief from a high-g_2 shock (since c_{2o} can now adjust), increasing μ_2^{Ramsey} relative to μ_2^{Reopt}
- The second term: $\tilde{\lambda}_1 > 0$ means taxing the old relaxes the implementability constraint; this term is positive, encouraging deviation

Time Inconsistency: Why Roth is Worse

$$M^* = \max_{\tilde{g}_2} \left[(\tilde{\mu}_2^{\text{Ramsey}} - \tilde{\mu}_2^{\text{Reopt}}) \tilde{g}_2 + \tilde{\lambda}_1 \tilde{c}_{2o}^{\text{Ramsey}} \right]$$

- Maximand is increasing in g_2
- Under **Roth**:
  - The Ramsey plan sets $\tau_2^s = 0$ and keeps c_{2o} constant
  - Second term of M^* independent of g_2
- Under **traditional 401k**:
  - The Ramsey plan uses τ_2^s to spread risk $\Rightarrow c_{2o}$ decreases with g_2
  - The gain $\tilde{\lambda}_1 \tilde{c}_{2o}^{\text{Ramsey}}$ is decreasing in $\tilde{g}_2$, partially offsetting first term
  - $\tilde{\mu}_2^{\text{Ramsey}}$ grows with g_2 at a slower rate, so first term is also growing at a slower rate
- $M^*_{\text{Roth}} > M^*_{\text{reg}}$

Richer Model: Households

- Overlapping-generations model; each household lives for N periods
  - Work for ages 1 through H; retire at age $H+1$
  - Discount factor $\beta < 1$; flow utility $u(c,n) = \frac{c^{1-\sigma}}{1-\sigma} - \psi \frac{n^{1+\xi}}{1+\xi}$
- Population grows at rate n
- At birth, draw permanent productivity type $j \in \{1, ..., M\}$
  - Within type, deterministic age-productivity profile $z_{j1}, ..., z_{jH}$

Richer Model: Technology and Government

- Wage on effective labor grows at rate γ
- Constant rate of return on capital r
- Government taxes: labor income (τ^{ℓ}), capital income (τ^{k}), pays Social Security ($p_j \gamma^t$)

Richer Model: Saving Options

- Taxable account (balance a): capital income taxed at rate τ^k
- Retirement account (balance d): subject to contribution limits $\bar{s}\gamma^t$ (or labor income, whichever is smaller)

Richer Model: Government Budget Constraint

- Government budget constraint (Roth economy):

$$B_t = (1+r)B_{t-1} + \gamma^t \sum_{i=t-N+1}^{t-H} \sum_{j=1}^{M} \pi_j p_j$$
$$-\tau^\ell \gamma^t w \sum_{i=t-H+1}^{t} \sum_{j=1}^{M} \pi_j z_{j,t-i+1} \ell_{tj}^i - \tau^k r \sum_{i=t-N+1}^{t} \sum_{j=1}^{M} \pi_j a_{j,t-i+1}$$

- Government budget constraint (traditional economy):

$$B_t = (1+r)B_{t-1} + \gamma^t \sum_{i=t-N+1}^{t-H} \sum_{j=1}^{M} \pi_j p_j$$
$$-\tau^\ell \sum_{i=t-N+1}^{t} \sum_{j=1}^{M} \pi_j [\gamma^t w z_{t-i+1,j} \ell_{tj}^i - (d_{tj}^i - (1+r) d_{t-1,j}^i)]$$
$$-\tau^k r \sum_{i=t-N+1}^{t} \sum_{j=1}^{M} \pi_j a_{j,t-i+1}$$

Political Process

- Period 0: households vote on constant tax rates τ^{ℓ}, τ^{k}, subject to present-value budget constraint of the government
- Conjecture: a Concorcet winner exists (if not, use probabilistic voting)
- Period $\hat{t}$: MIT shock to government spending. Households vote on new constant tax rates, and whether to default on past promises and tax retirement accounts
- Trigger: if retirement accounts are taxed, they disappear for ever

Algorithm: Initial Vote

- Solve value functions, policy functions for each value of τ^ℓ, τ^k (fast: 2 endogenous state variables, age, permanent type)
- Choose exogenous initial distribution of taxable and nontaxable assets by age
- For each (τ^ℓ, τ^k), compute present-value gov't budget balance going forward, retain pairs that make the budget constraint work (interpolating as needed)
- Using value functions, evaluate outcome of vote over admissible pairs
- Optional step: Iterate so initial distribution is invariant distribution according to the equilibrium pair

Algorithm: Reoptimization with no Promise Default

- Initial distribution = invariant distribution from before shock
- Compute budget balance for each (τ^{ℓ}, τ^{k}) with shock
- Using value functions (same as before), evaluate outcome of vote over admissible pairs

Algorithm: Reoptimization with Default on Past Promises

- Compute value functions and policy functions if retirement accounts are treated as taxable
- Initial distribution = invariant distribution from before shock
- Compute budget balance for each (τ^{ℓ}, τ^{k}) with shock
- Using value functions, evaluate outcome of vote over admissible pairs
- Evaluate when defaulting on the promise not to tax wins a majority vote (as a function of β)

Purpose of Richer Model

- Get a quantitative evaluation of the two main channels:
  - Magnitude of improved risk sharing under traditional 401k
  - Minimum discount factor necessary to support no default on past promises under two scenarios

Summary and Next Steps

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  - The Roth promise is a commitment to incomplete markets: it prevents optimal risk sharing between generations
  - It also *worsens* the time-inconsistency problem: taxing accumulated Roth balances is tempting ex post
- **Analytical model**: provides clean theoretical results on both margins
- **Quantitative model** (in progress):
  - Calibrate to U.S. lifecycle earnings profiles, contribution limits, Social Security
  - Compute voting equilibria before and after a spending shock
  - Measure welfare losses and degree of commitment required
- **Policy implication**: The ongoing expansion of Roth (SECURE 2.0 mandated Roth catch-up contributions; growing Roth 401(k) participation) increases the political economy risk of future taxation of these accounts